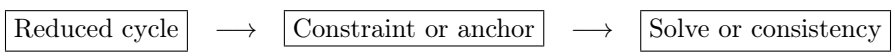
Signed graph cycles, anchors and constraints

2026-11-01. Ten main declarations, eight controls and eighteen named axiom readbacks passed.

Derived constraint semantics. From $a + bk = 0$ and $b \neq 0$, the native expression $(-a)b^{-1}$ equals k. When $b = 0$, the relation forces $a = 0$, matching the native consistency assertion rather than inventing a solve.

Same-root branches. A reduced edge is $c_u R + c_v R = A + Bk$. Opposite signs derive the source equation $-A + (-B)k = 0$ and its returned quotient equals k. Equal signs derive $R = A/(2c) + B/(2c)k$ under $2 \neq 0, c \neq 0$. Explicit sign-domain classification and the conditional native denominator 2 versus -2 are checked.



Existing fixed anchor. Two valid descriptions $R = x + yk = a + bk$ yield $(x-a)+(y-b)k = 0$. Nonzero target coefficient gives k; zero target coefficient forces equal constants. A first anchor only records a root description. Metadata mutation is a separate obligation.

Controls. Direct solve, positive-cycle signs, both negative-cycle signs, existing-anchor solve, equal-anchor consistency, inconsistent zero coefficient rejection and characteristic-two boundary passed. For prime modulus $r > 2$, the field has nonzero2; the native representative $r - 2$ and inversion routine still need their word-level bridge.

Failures retained. Initial negation orientation and redundant tactic failed; a later extension lacked DecidableEq for the conditional denominator. Corrected proof and controls pass. Failed sources/logs remain; nonfatal warnings retained.

Boundaries. Input relations and root equations must come from graph invariants. This packet does not prove forest mutation, canonical words, i8 sign preservation, modular inverse implementation, index/counter safety or compiled equivalence. Prior add/sub/mul correspondence is not an inverse theorem. Independent semantic review pending. Fresh checks retain cached dependency trust boundaries. No scientific execution or performance claim.

Source: three_leads.rs SignedGraph edge/anchor/constraint. Artifacts: REPORT.md, status.json, receipts, exact input bindings, source snapshot and editable flow.svg.